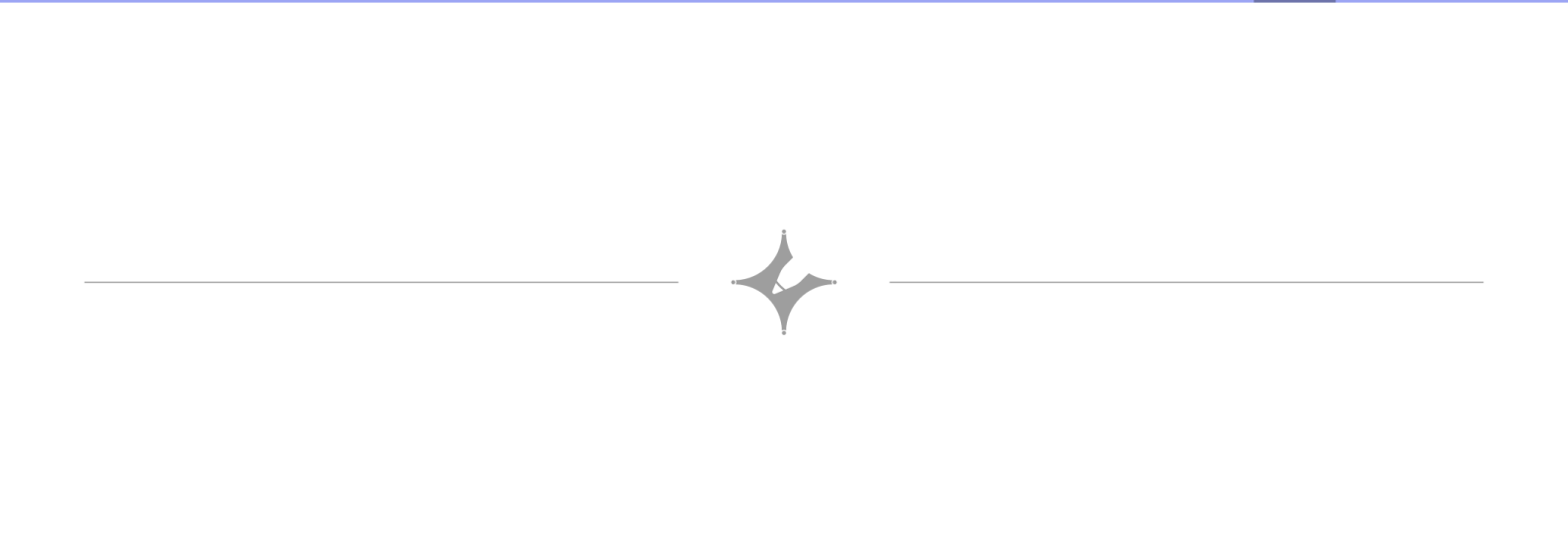
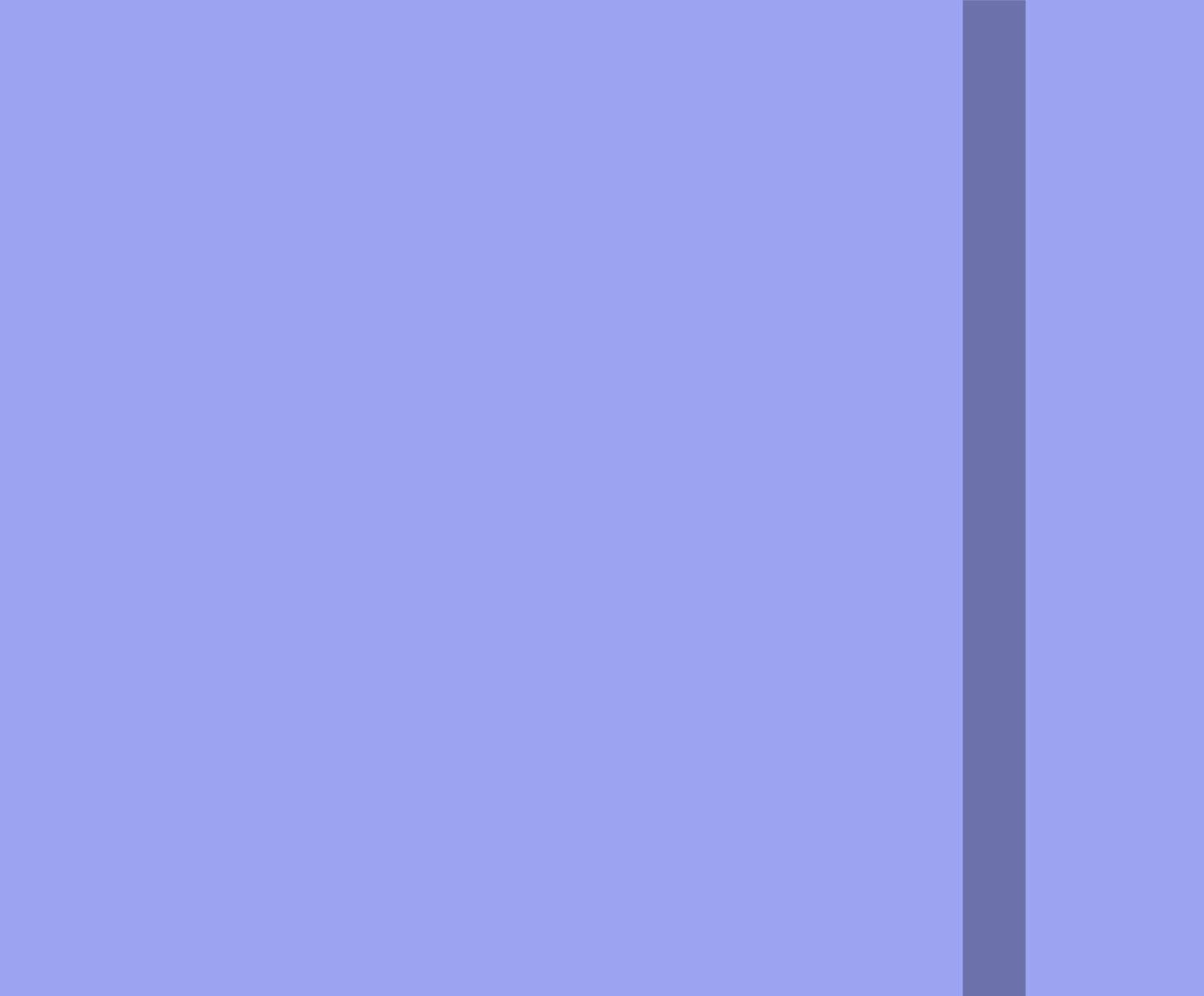




Pump Theorem.

Let L Regular. $\exists p \geq 1$, s.t. $\forall w \in L$ with $|w| \geq p$

可被分为3部分 $w = xyz$, 满足

(1). $\forall i \geq 0, xy^i z \in L$

(2). $|y| > 0$

(3). $|xy| \leq p$.

pumping length

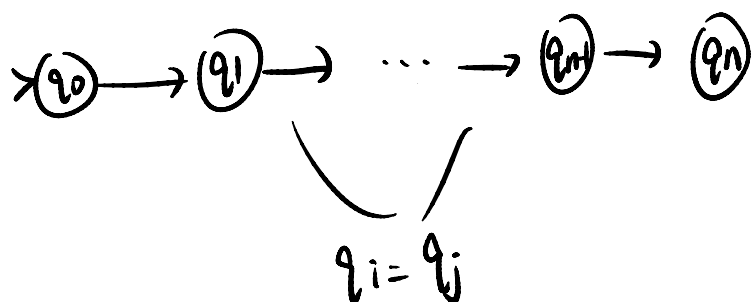
例: ab^*a . $p=2 : aa \quad \times$
 $p=3 : aba \quad \checkmark$

下证 pump theorem.

$\exists$ DFA M decide L .

令 $p = \# \text{ states of } M$

若 $w \in L$ with $|w| \geq p$.



$n \geq p$.

$\therefore$ 必有 $0 \leq i < j \leq p$, s.t. $q_i = q_j$. (抽屉原理)

令前、中、后为 $x y z$, 显然 $x y^i z \in L, \forall i$ ①

$$|y| = j - i > 0 \quad \text{②.}$$

$$|xy| = j \leq p \quad \text{③.}$$

#.

例: $0^n 1^n$ 非正则 (记为 A)

设 A 正则

$\therefore$ let p 为 pump length.

let $w = 0^p 1^p$, $w = xyz$ 满足3个条件.

由(2)、(3), $y = 0^k$, $x y^p z = 0^{p-k} 1^p \notin A$

矛盾 $0^n 1^n$ 非正则.

#.

(lec1)

Chapter 5 : 自动机 (续)

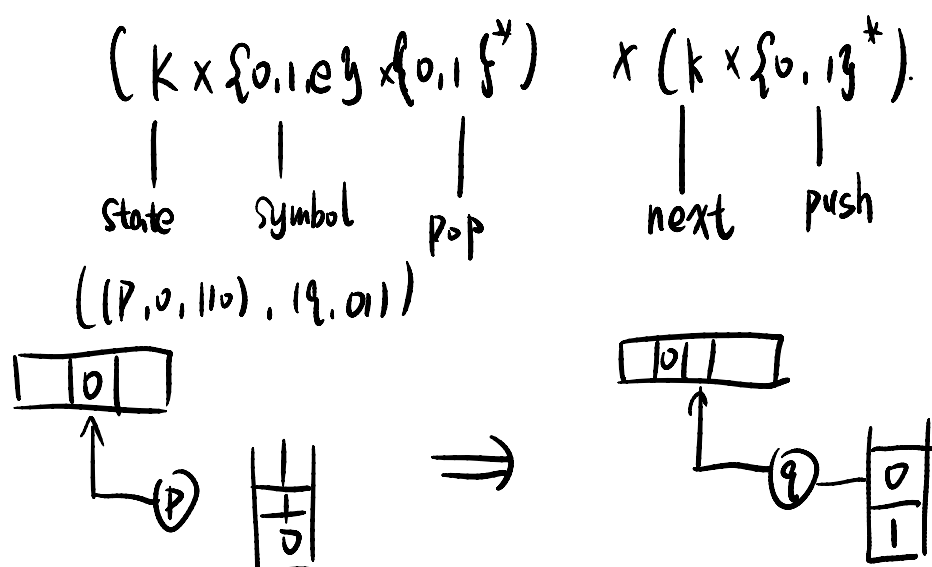
定义:

pushdown Automaton (PDA)

$PDA = NFA + Stack$

PDA is a 4-tuple (K, Δ, S, F)
state initial accepting

Δ : transition relation.



定义 2: Configuration state + 从读写头往后 + stack.

a member of $K \times \{0,1\}^* \times \{0,1\}^*$.

$(p, \alpha, \beta) \quad \Gamma_p (q, \gamma, \beta)$

↘ yields in 1 step

$\Gamma_p^* \rightarrow \text{yields}$

P accepts $w \in \{0,1\}^*$ if.

$(s, w, e) \Gamma_p^* (q, e, e)$

接受

read
end

栈空

P decides $L(P)$.
Context-free (定义).

例: $L = \{w \in \{0,1\}^*; \#0 = \#1\}$

$K = F = \{q\}, S = q.$

$\Delta = \{ (q, 0, e), (q, 0), ((q, 0, 1), (q, e)), ((q, 1, e), (q, 1)), ((q, 1, 0), (q, e)) \}.$

例: $L = \{w w^{\text{Reverse}} : w \in \{0,1\}^*\}.$

$K = \{l, r\}$

$S = ($ $\Delta = \{ ((l, 0, e), (e, 0)), ((l, 1, e), (l, 1))$

$F = \{r\}$

$((l, e, e), (l, e))$
 $((r, 0, 0), (r, e)), ((r, 1, 1), (r, e)).$

language gen: produce $w \in L$

$\downarrow$
 Grammar.

Context-free Grammar. (CFG).

$S \rightarrow 0S1$

$S \rightarrow A$

$A \rightarrow 0$

$A \rightarrow e$

} rule + non-terminal + Start Symbol + terminal
 $S, A.$ S $0, 1$

定义:

一个 CFG $G = (V, \Sigma, R)$

V : finite set symbols, 包含 0, 1.

$S \in V \setminus \{0, 1\}$: Start Symbol

$R \subseteq (V \setminus \{0, 1\}) \times V^*$ (finite).

$(A, u), A \rightarrow u.$

定义: derivation

for $x, y, u \in V^*, A \in V \setminus \{0, 1\}.$

$xAy \Rightarrow_G xuy$ if $(A, u) \in R$

$w \Rightarrow_G^* u$

↓
derive

G generate $w \in \{0, 1\}^*$ if $S \Rightarrow_G^* w.$

G generate $L(G).$

例: $\{w \in \{0, 1\}^* : w = w^R\}.$

$S \rightarrow \epsilon$

$S \rightarrow 0S0$

$S \rightarrow 1$

$S \rightarrow 1S1$

$S \rightarrow 1$

定理: $PDA \Leftrightarrow CFG$

证: $\Leftarrow$

Idea:

1. in stack, 用CFG生成一个G
 2. 对比G. 输入
 3. match?
- 边变-边匹配.

(构造略. 比较 dirty work).

$\Rightarrow$: 先化简PDA

定义: Simple PDA

A PDA simple:

(1). $|F| = 1$

(2). $((p, a, d), (q, \beta)) \in A$

either $a = \epsilon, |\beta| = 1$

or $|\alpha| = 1, \beta = \epsilon$.

引理: $PDA \Rightarrow \text{simple PDA}$.

(1). if $|F| > 1$. 建新状态.

(2). 2.1. $|\alpha| \geq 1, |\beta| \geq 1$. 拆成 2.2 与 2.3

2.2. $|\alpha| > 1, \beta = \epsilon$ 一个-个

2.3. $|\beta| > 1, \alpha = e$ 一个 - 一个

2.4: $\alpha = \beta = e$ push + pop

继续定理证明, 要 Simple PDA $\Rightarrow$ CFG

$$PDA = (K, \Sigma, \Gamma, \delta, q_0, \{f\}).$$

$$G = (V, S, R).$$

Goal: $A_{PDA} \xRightarrow{*}_G w$ for some $w \in \{0,1\}^*$ $\Leftrightarrow w \in \{u \mid \{0,1\}^*\}$.

$$(p, u) \vdash_p^* (q, e, e)$$

$$S = A s f$$

$$A p p \rightarrow e, A p q \rightarrow A p r A r q$$

$$\forall ((p, a, e), (p', a)), ((q', b, a), (q, e)) \in \Delta$$

$$A p q \rightarrow a A p' q' b$$

思路类似区间 dp.

(lec 8)

Chapter 6 : 图灵机

PDA 无法判定 $\{0^n 1^n 0^n\}$, 因此考虑更强的模型.

定义1: Turing Machine.

$$M = (K, \Sigma, \Gamma, \delta)$$

K : finite set of states

s_{ek} : Initial state

Σ : finite set of symbol $\geq \{0, 1, \triangleright, \sqcup\}$

δ : transition function

$$K \times \Sigma \longrightarrow K \times \Sigma \times \{L, R, S, H\}$$

On input $x = x_0 x_1 \dots x_{n-1}$

T

$\triangleright$	x_0	x_1
------------------	-------	-------

 $\dots$

x_{n-1}

$$T[0] = 0, \quad T[i] = x_{i-1}, \quad i = 0, S = 0.$$

repeat.

1. $(s', a, D) \leftarrow \delta(s, T[i])$

2. $T[i] \leftarrow a, \quad S \leftarrow s'$

3. if $D == R$
 $i = i + 1$

4. if $D == L$
 $i = i - 1$

5. if $D == H$
 halt.

if halt

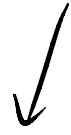
Output $M(n) = T[0] \dots T[i]$, $T[i+1]$ 为第一个空格

else

Output $M(x) = 1$

A TM M compute a function $f: \{0,1\}^* \rightarrow \{0,1\}^*$

if $\forall x \in \{0,1\}^*, M(x) = f(x)$



Computable function

NAND-TM programming language

NAND-TM = NAND-CIRC + arrays + loops

1. i : integer value

Other: bool value

2. scalars: bool variable

arrays: index by i (共用)

3. input X , output Y 为 array

4. MOD AND JUMP: modify i , jump to first line

if $a = b = 1$

$i \leftarrow i + 1$

if $a = 0, b = 1$

$i \leftarrow i - 1$

if $a = 1, b = 0$

$i \leftarrow i$
 if $a = b = 0$
 halt

5. default value: 0, 除了 χ .

定理1: $TM \Leftrightarrow NAND-TM$.

idea: states $\longleftrightarrow$ scalar variable
 tape $\longleftrightarrow$ array
 head $\longleftrightarrow i$

$\Rightarrow$:

$M = (K, \Sigma, s, \delta)$

k states $\rightarrow \lceil \log k \rceil$ scalar variables

Σ $\xrightarrow{\text{tape}}$ $\lceil \log |\Sigma| \rceil$ arrays

$\{L, R, S, H\} \rightarrow \{01, 10, 11, 00\}$

$\delta: K \times \Sigma \rightarrow K \times \Sigma \times \{L, R, S, H\}$.

$\downarrow$
 $\delta: \{0, 1\}^{\lceil \log k \rceil + \lceil \log \Sigma \rceil} \rightarrow \{0, 1\}^{\lceil \log k \rceil + \lceil \log \Sigma \rceil + 2}$

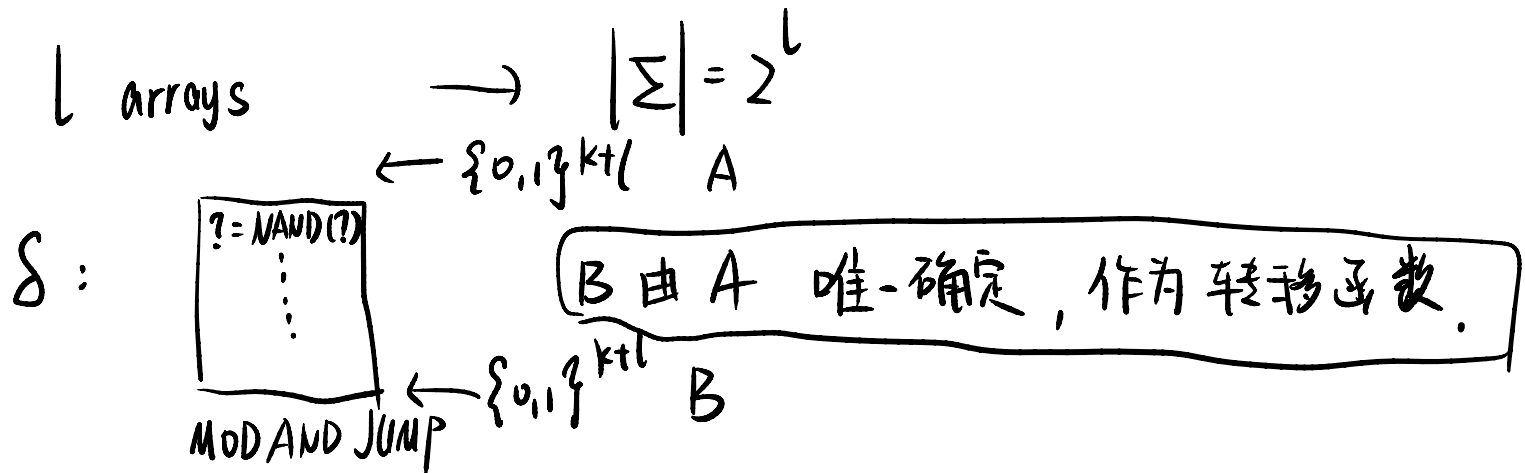
finite function

$\downarrow$

$\exists$ NAND-CIRC 计算此函数.

NAND-TM $\Rightarrow$ Turing Machine

k scalar variables $\rightarrow 2^k$ states



#.

NAND-TM 语法糖.

GOTO. 1 ? = NAND(?,?) $\vdots$ t ? = NAND(?,?) t+1 MOD AND JUMP	$\Leftrightarrow$	if line = 1 () line = 2 $\vdots$ if line = t (). MOD AND JUMP
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GOTO: 改变 line 的值即可

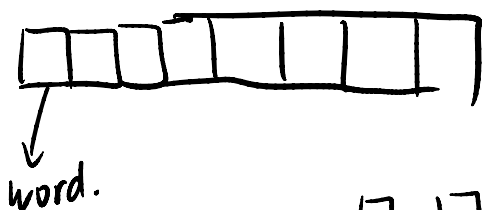
$\Downarrow$

"loop"

高维数组 i, j . (拉平成一维数组即可).

NAND-RAM

RAM.



□ □ □ Registers

1. read/write memory (Random access)
2. Register 操作
3. flow control

NAND-RAM program

1. integer variables (bounded)
2. index access array
3. 四则运算

定理 2: $NAND-TM \Leftrightarrow NAND-RAM$

(证明略).

